

Maths Trail at the Dundee Botanic Gardens

Maths Week Scotland - Dundee University

Welcome! This site was created in 2026 at the University of Dundee for our Maths Trail at the Dundee Botanic Gardens, as part of the wider Maths Week Scotland programme of events. It is designed to be used while exploring the Botanic Gardens and using a smart-phone to scan QR barcodes taking you to individual pages here.

The Analemma Project

Artist Lucy Smith created the Analemma Project sculpture. She completed her PhD “Making the Conceptual Tangible: The Role of Art in Communicating Science” here at the University of Dundee.

The Analemma is an unequal figure-of-eight shape, created by tracking the sun’s position in the sky at the same time each day over a period of one year.

Why doesn’t the Earth fall onto the Sun?

The Analemma traces that figure-of-eight because Earth’s orbit isn’t a perfect circle travelled at constant speed – it’s an ellipse, and Earth moves faster when it’s closer to the Sun. Working out exactly *why* orbits look like this took four scientists and the better part of three centuries. Here’s that story in four small experiments, each one playable on your phone: **Galileo**, who found the first hard evidence that not everything orbits the Earth; **Kepler**, who worked out the precise shape of an orbit from observation; **Newton**, who explained *why* that shape occurs, from a single law of gravity; and **Einstein**, whose refinement of Newton’s picture explains a tiny mismatch even Newton’s theory could not.

Bird Song

Listen to the birds!

Circadian Rhythm

If you were to watch the water lily here through a full day you would see its flowers open in the morning and close again in the evening.

However, if it was left in a room with the lights left on permanently, it would keep opening and closing on roughly the same rhythm for a while, gradually drifting out of step with real time. The water lily isn't just reacting to light like a switch, it has its own internal molecular clock that is entrained to day-night cycle.

Your cells also have an internal molecular clock. And if you've ever experienced jet lag, your internal molecular clock has become out-of-sync with your new day-night cycle. It takes a number of days of light stimulus in your new environment before your body clock syncs with its environment.

Why leaves crinkle: differential growth in a disc

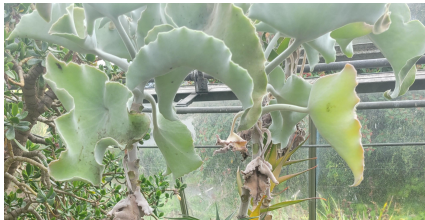
Why are some leaves more wrinkled than others? And what causes this to happen?

Consider a circular disc of radius, R .

The perimeter is $2\pi R$ whilst the area of the circle is πR^2 . It can be shown that this circle is the most efficient way to contain the area πR^2 .

Now imagine distorting the disc so that the circumference is wrinkled. The circumference of the wrinkled shape is larger than $2\pi R$.

In the plant at this stop, notice that the leaves are wrinkled rather than smooth. Why does evolution find this shape for this particular plant and how does it happen?



Fibonacci Numbers

Look at a pinecone from the bottom. How many spirals go in the clockwise direction (green lines in the photo)? How many spirals go in the counter-clockwise direction (yellow lines)? Wouldn't you expect that they would be the same?

Image from the MENSA for Kids that shows Fibonacci numbers in pinecone:

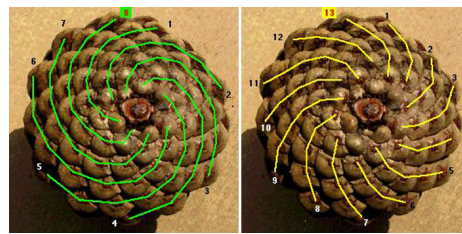


Figure 1: Fibonacci Numbers in Pinecone. There are 8 spirals in the clockwise direction and 13 spirals in the counter-clockwise direction.

The number of spirals found on pinecones are almost always Fibonacci numbers, which are .

Fractal

A fractal is a structure that possessed the property of **self-similarity**. Specifically, it refers to a structure where the exact same shape repeatedly appears when the entire figure is broken down into several smaller parts.

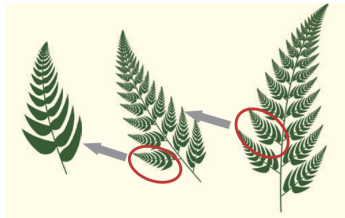


Figure 1: Fractal pattern in fern leave

While fractal patterns and shapes appear complex at first glance, they are actually the result of repeating simple rules and operations.

Helicopter Seeds

The seeds from maple, elm and ash trees have curved wings to let them stay in the air longer. The wind will pick up these seeds and carry them to suitable ground. These seeds twirl and spin around. They will then drop and sprout a new tree. Some people call these seeds **helicopter seeds** because they look like and can spin like a propeller.

Here you will learn how to make your own spinner.

1. Fold the middle flap in the opposite direction of the other two flaps. Put the paper clip on the end of the middle flap.
2. Find a maple seed spinner. Drop both spinners from the same height and compare how they spin. Does your spinner work as well as nature's?

Seven Bridges of Königsberg

In the 1700s the city of Königsberg was built around two branches of a river, with an island sitting where they met. Seven bridges connected the four separate pieces of land – and the people of the city had a favourite Sunday puzzle: could you walk through the city and cross every one of the seven bridges exactly once?

Nobody could ever do it. Nobody could ever explain *why*, either – until a mathematician called Leonhard Euler looked at the problem in 1736, and realised something surprising: it doesn't matter how big the islands are, how long the bridges are, or what order you visit things in. All that matters is *which* pieces of land are joined to *which*, and by how many bridges. That single idea – throw away the shape and the distance, keep only the connections – was the birth of an entire branch of mathematics: graph theory, and later, topology.

Phyllotaxis

Plants arrange their leaves around a stem in regular patterns. The angle between one leaf and the next is called the divergence angle, and this pattern of leaf arrangement is known as phyllotaxis. Many different plant species share similar phyllotaxis patterns, and these can be described using a small number of mathematical patterns.

Which of the following leaf arrangement pattern do you find in this plant? Please record the number of each leaf arrangement pattern in the table provided on the Maths Trail map: **distichous** (180°), **Fibonacci spiral** (137.5°), **decussate** (90°), or **tricussate** (60°).

Image from the University of Tokyo that shows the classification patterns:

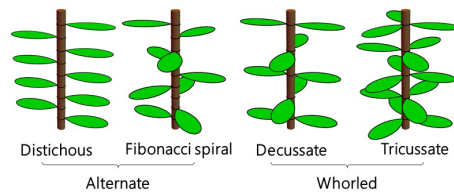


Figure 1: Phyllotactic Patterns

The Travelling Salesman problem

If you have followed the trail around this exhibition, your route is annotated on the map below.

Suppose you wanted to identify the shortest possible route from start to finish, i.e. how much further did you just walk than you needed to? This is the **Travelling Salesman Problem** (TSP) – one of the most famous problems in mathematics and computer science. It sounds simple: just try every order of stops and keep the shortest. The catch is *how many* orders there are – with just 20 stops, there are over 60 quadrillion, far too many to check by brute force. Try the app on the website to compare your own route against the optimal one.

Logarithmic Spiral

Logarithmic spirals in plants are a striking example of geometry in nature, where botanical structures grow according to the Golden Ratio and the Fibonacci sequence.

Spiralling Succulents

Many varieties of succulents arrange their leaves outward from a central point along an expanding logarithmic trajectory to ensure uniform access to sunlight and water collection.

Image from the Fractal Dimensions in Circular and Spiral Phenomena that shows logarithmic spiral in Aloe Polyphylla:



Figure 1: Logarithmic spiral in Aloe Polyphylla

Symmetry in Plants

Links to Group Theory.

Estimating the Height of a Tree using Trigonometry



Suppose you need to measure the height of the tree in front of you, standing on the ground with nothing but your own two feet and the phone that you are using on this tour.

The idea

Everything below comes down to one right-angled triangle:

- your eye,
- the point on the tree that is at the same height as your eye,
- the top of the tree.

End of Maths Trail Questionnaire

Display at the end of Maths Trail (near the reception?). Links to Microsoft form, which asks feedback and also asks to enter the counted numbers for Phyllotaxis patterns.